PSYC 2012: Biological Psychology Week 1: Intro to Biological Psychology & Anatomy of the Nervous System

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Goal of This Week’s Lecture:

To learn what biological psychology is and how it is studied, and to learn the basic anatomical organization of the nervous system

What is Biological Psychology?

Definition of Psychology:

Definition of Psychology:

The scientific study of the human mind and its functions, esp. thoughts, feelings, and behavior

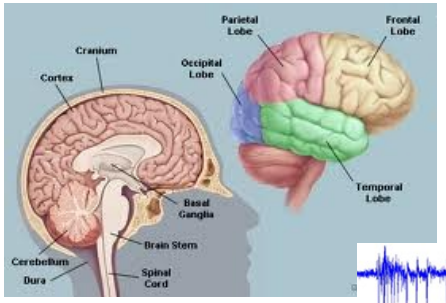
Definition of <u>Biological</u> Psychology:

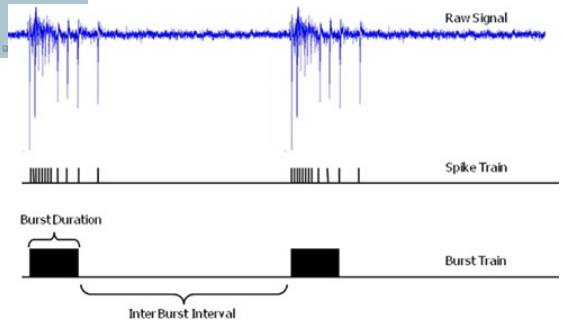
Definition of <u>Biological</u> Psychology:

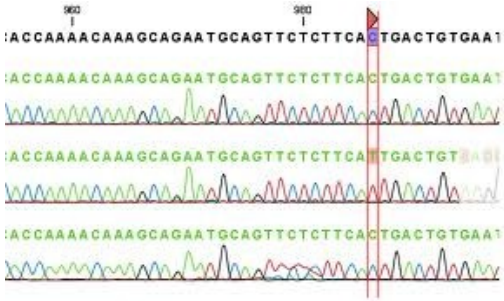
The application of the principles of biology to the study of thoughts, feelings, and behavior in humans and animals

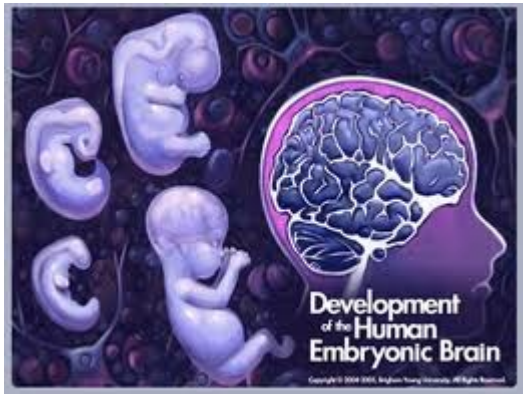
Principles of Biology

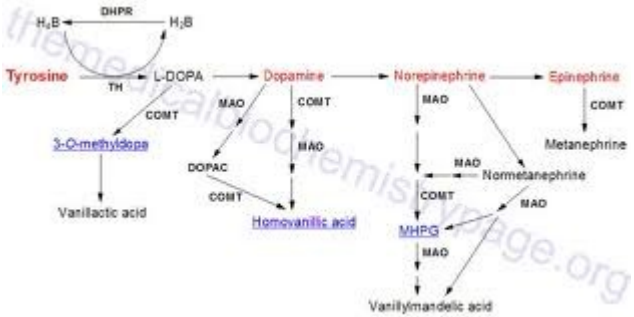
- **Anatomy**—the study of the different organs and structures
- **Cellular anatomy or cell biology**—the study of cell organelles and structures
- **Physiology**—the study of how living systems function
- **Genetics**—the study of how genes and evolution affect biological processes
- **Developmental biology**—the study of how an organism and its tissues are created during development
- **Biochemistry**—the study of molecules of life and how they influence living systems











Key Point:

Biological psychology focuses on the role genes, molecules, cells, tissues and organs play in cognitive, affective, and behavioral processes

The Study of Biological Psychology

Scientific Research

Scientific Research:

Systematic, controlled, and critical examination of the world, often using experiments to explore the relationship between two variables

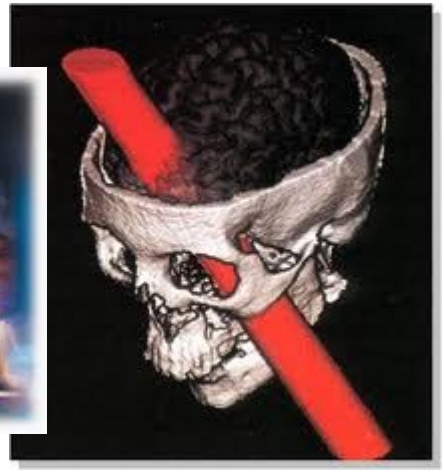
Experiments

- Subjects are placed in one of two conditions in which one variable is manipulated

- The effect of manipulating that variable on some outcome variable(s) that can be measured is determined

5 Types of Biopsychological Variables That Can Be Manipulated

- **Genetic manipulations**— selective breeding, transgenic manipulation (animals only)
- **Neuropharmacology**—giving chemicals that influence brain function to see how it affects behavior (animals and humans)
- **Electrical stimulation**—studying how increasing the activity of a brain region by stimulating it electrically influences behavior (animals and humans)
- **Trauma studies**—studying how damage to a brain region due to accidental injury affects behavior (humans)
- **Lesion studies**—creating intentional damage to brain regions to see how it affects behavior; can be temporary or permanent (animals only)



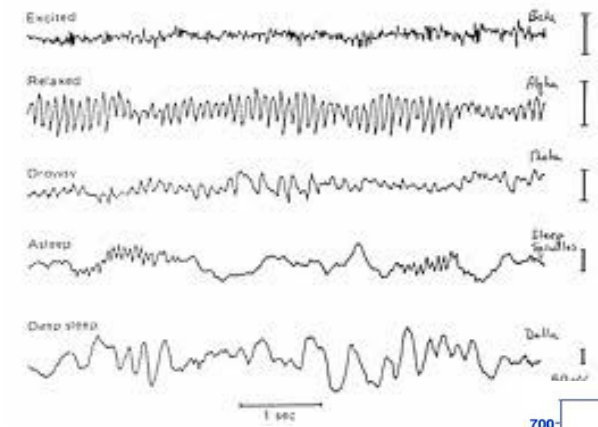
4 Types of Biopsychological Outcome Variables That Can Be Measured

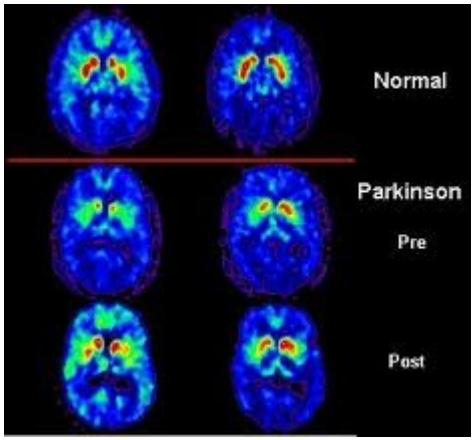
- **Neuroimaging**—non-invasive ways to visualize brain structures or their activity--MRI, fMRI, PET

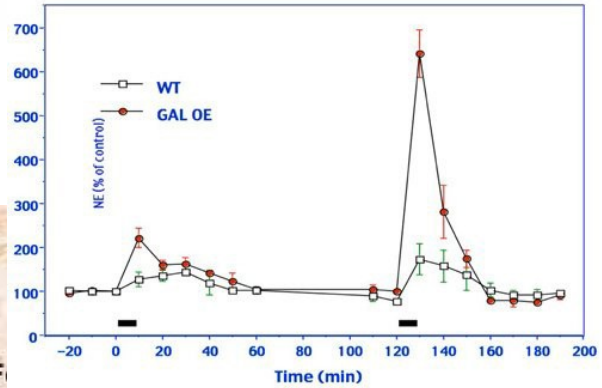
- **Electrophysiology**—measuring the electrical activity of the brain--EEG, EMG, patch clamping

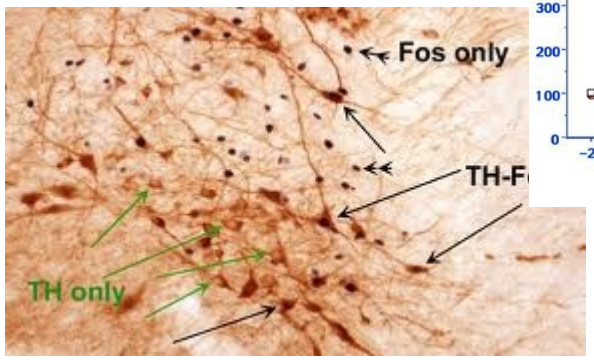
- **Neurochemistry**—measuring neurotransmitters, metabolites, signaling molecules in the nervous system

- **Neurocytology**—staining cells in the brain to explore the size and number of neurons or the connectivity of neural circuits



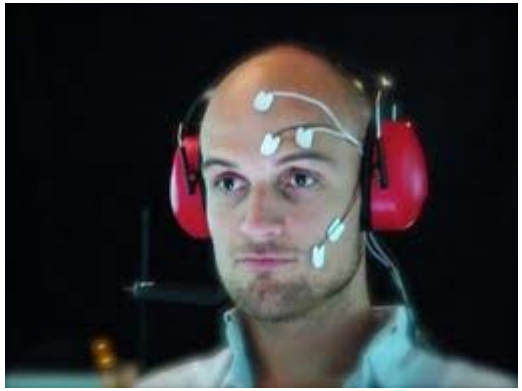






Human vs. Non-Human Subjects

- Animals are often used to study biological psychology



What are Some of the Pros and Cons of Animal Experimentation?

Pros

Cons

Pros and Cons of Animal Experimentation

Pros

- Share many of the same basic psychobiological pathways
- Ethically allowed to do things that aren’t allowed in humans
- Inexpensive way to get large numbers of subjects
- Less complex system

Cons

Pros and Cons of Animal Experimentation

Pros

- Share many of the same basic psychobiological pathways
- Ethically allowed to do things that aren't allowed in humans
- Inexpensive way to get large numbers of data points
- Less complex system

Cons

- Not all processes are the same between animals and humans—cerebral cortex
- Still ethically distasteful for people
- Still require approval from proper authorities
- Animals can't follow instructions or report their experiences to us

Course Goal:

To understand the principles of biological psychology and the fundamentals of the relationships between brain structure, brain function, and behavior

Anatomy of the Nervous System

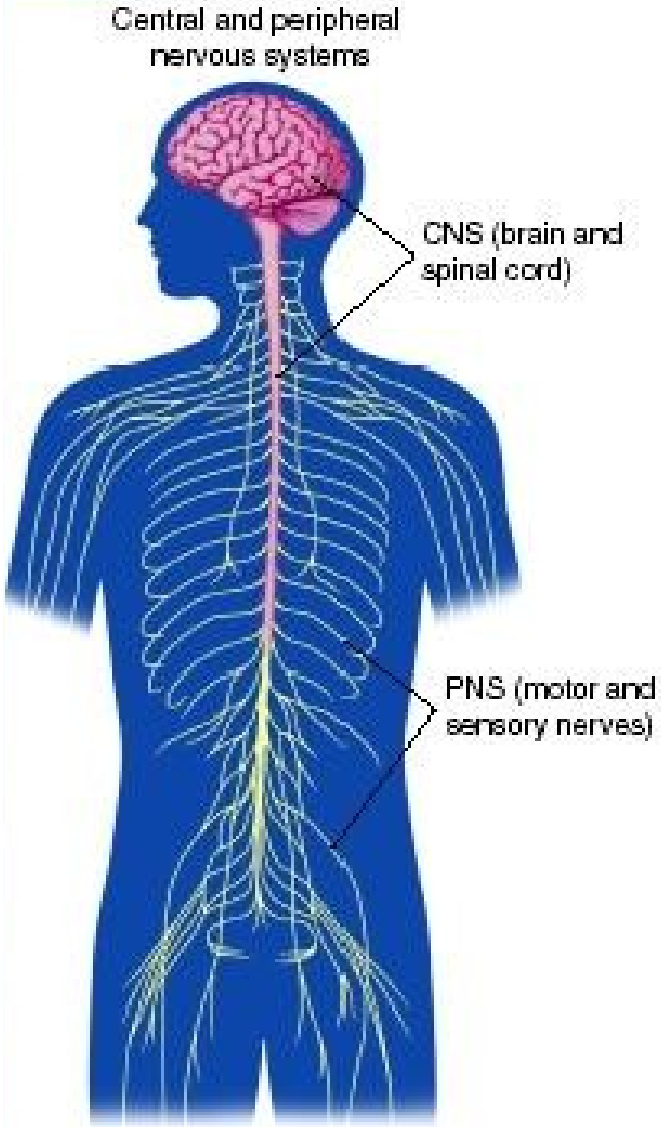
Nervous System:

The organs, tissues and cellular structures that are involved in the function of communication via the creation and transmission of electrical signals

Peripheral vs. Central Nervous System

- **Peripheral nervous system**—components of the nervous system outside the skull and spine, i.e., <u>the nerves</u>

- **Central nervous system**—components of the nervous system located in the skull and spine, i.e., the <u>brain and spinal cord</u>

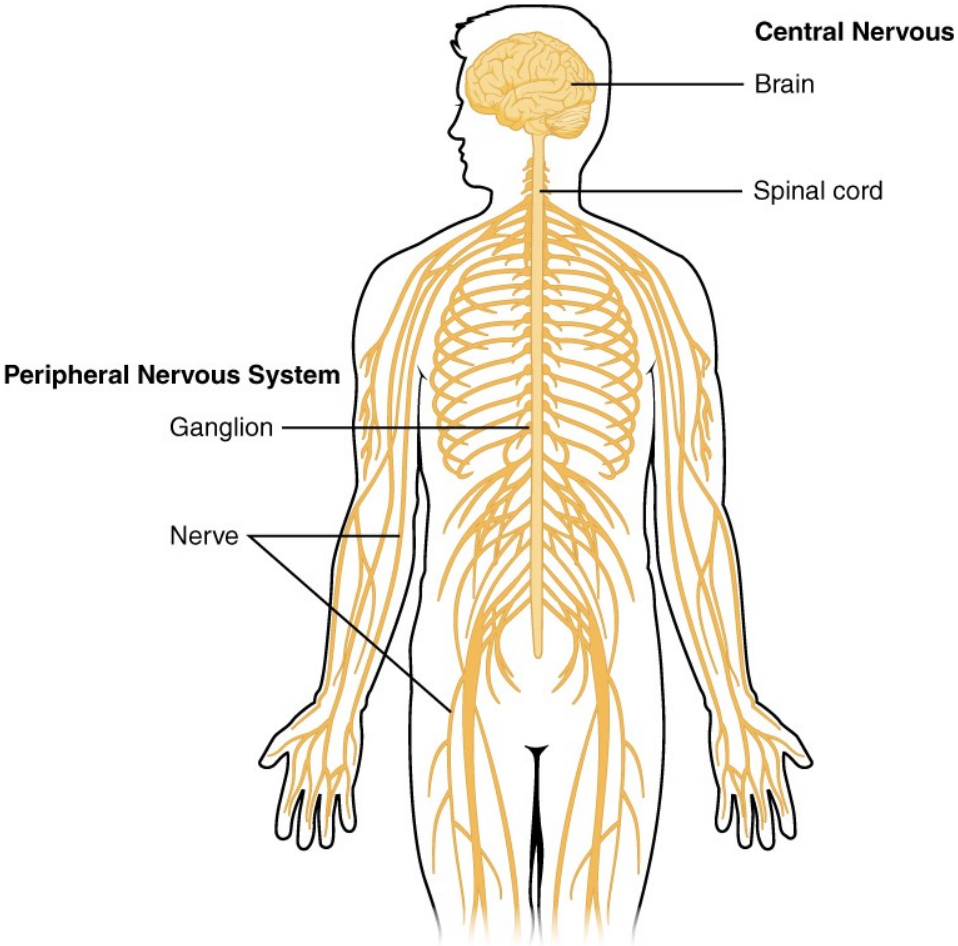


The Peripheral Nervous System

2 Main Components of the Peripheral Nervous System

- **Nerves:** enclosed, cable-like bundles that transmit electrical impulses to and from body structures and the brain

- **Ganglia**: clusters of nerve cell bodies associated with nerves

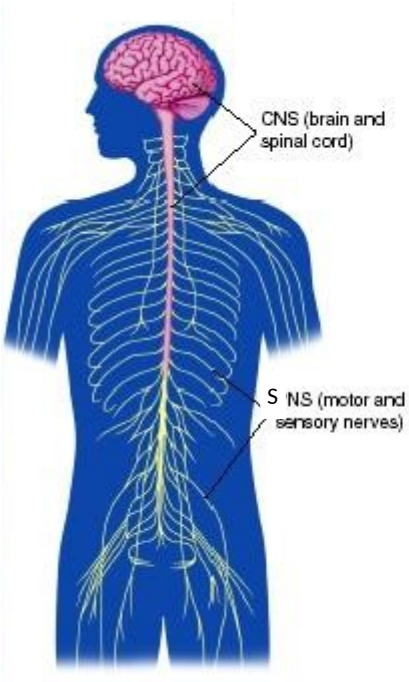


2 Main Branches of the Peripheral Nervous System

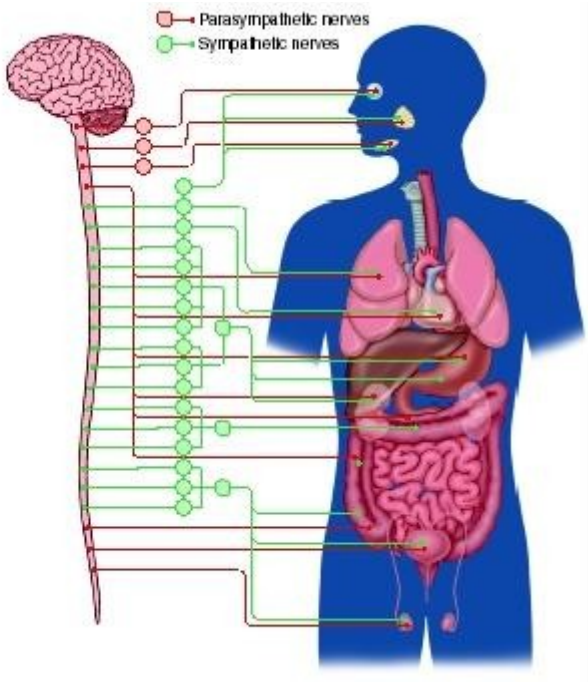
- **Somatic nervous system**—nerves carrying sensory information to the CNS and motor commands from the CNS

- **Autonomic nervous system**—special nerves that regulate activity of the internal organs & other involuntary activities

Somatic



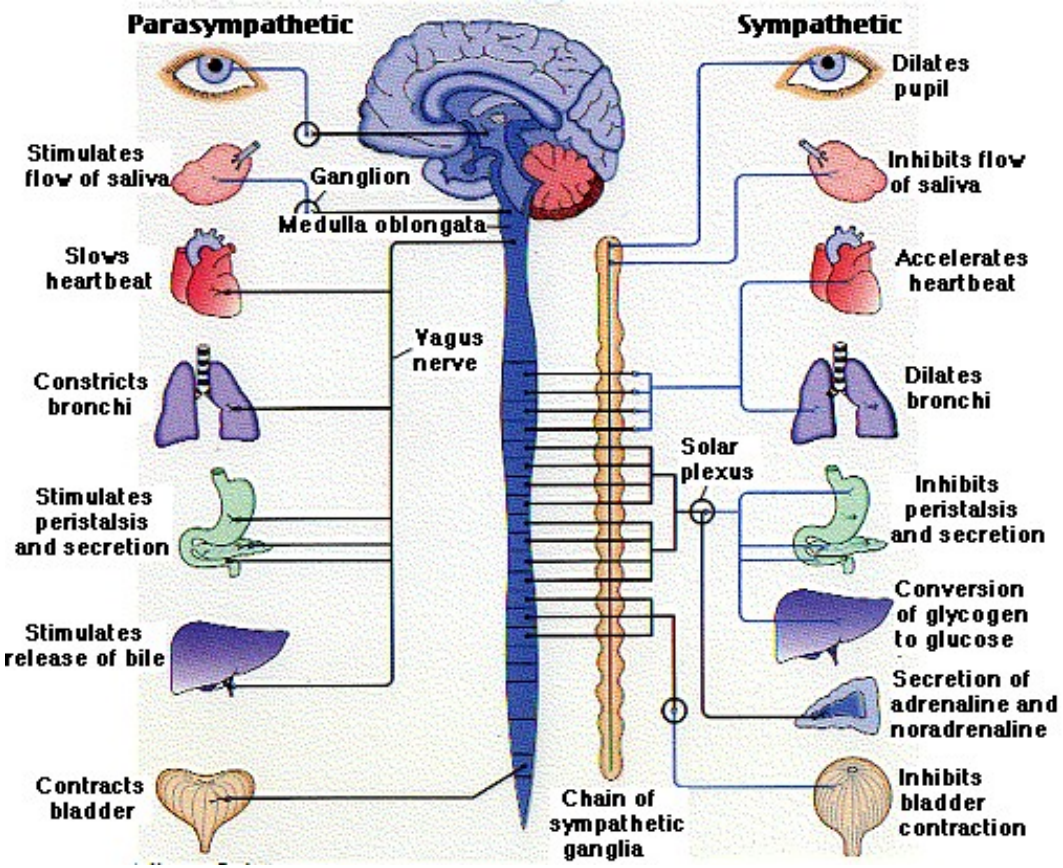
Autonomic



Autonomic Nervous System

- **Parasympathetic nervous system**—regulates day-to-day functions like digestion

- **Sympathetic nervous system**—regulates "fight or flight" response to threats and stressors



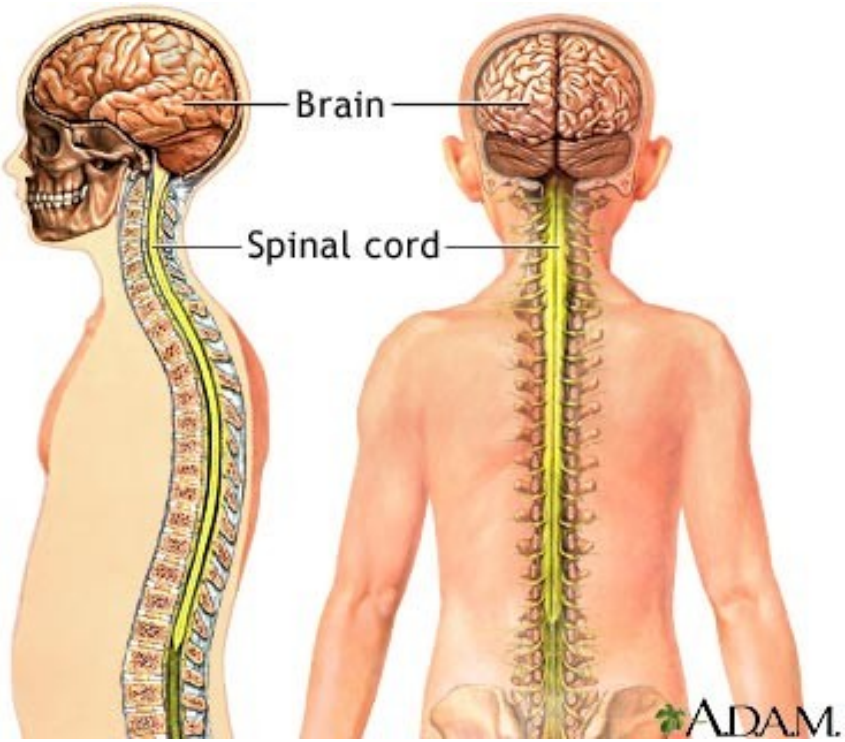
The Central Nervous System

The spinal cord and the brain

The Spinal Cord

- Functions primarily in transmission of signals to and from the brain and the rest of the body

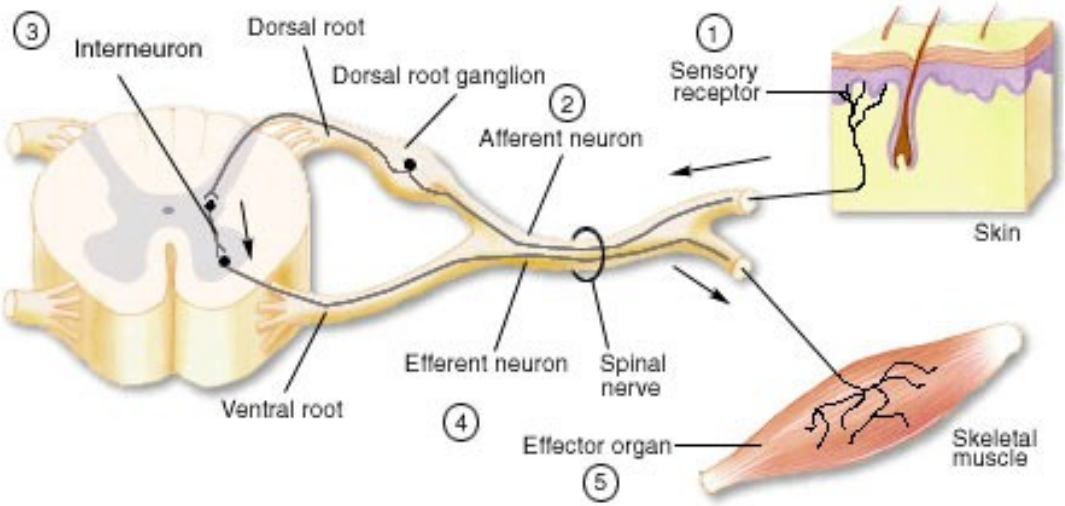
- Also acts independently of the brain with local reflexes and other neural circuits



The Spinal Cord & Spinal Nerves

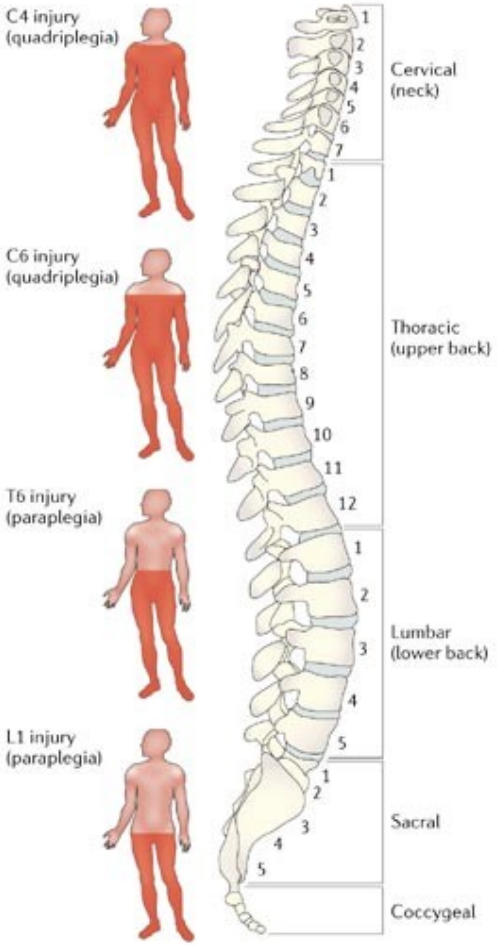
- **Dorsal roots**—bring sensory information to the CNS from the body—**afferent pathway**—damage to these causes **numbness** or loss of sensory information but not paralysis at the affected level

- **Ventral roots**—take motor commands from CNS to muscles—**efferent pathway**—damage to these causes **paralysis** or loss of motor control but not numbness at the affected level



The Spinal Cord and Neck Injury

- Physical trauma to the spine can sever communication between the brain and the rest of the body
- Causes both numbness <u>and</u> paralysis
- The higher up an injury occurs, the more of the body is affected
- Primary causes—car accidents, falls, sports injuries, assaults
- >12,000 people experience spinal cord injury each year



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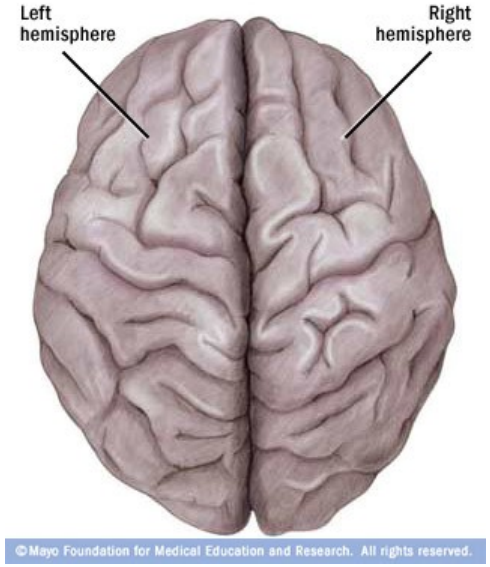
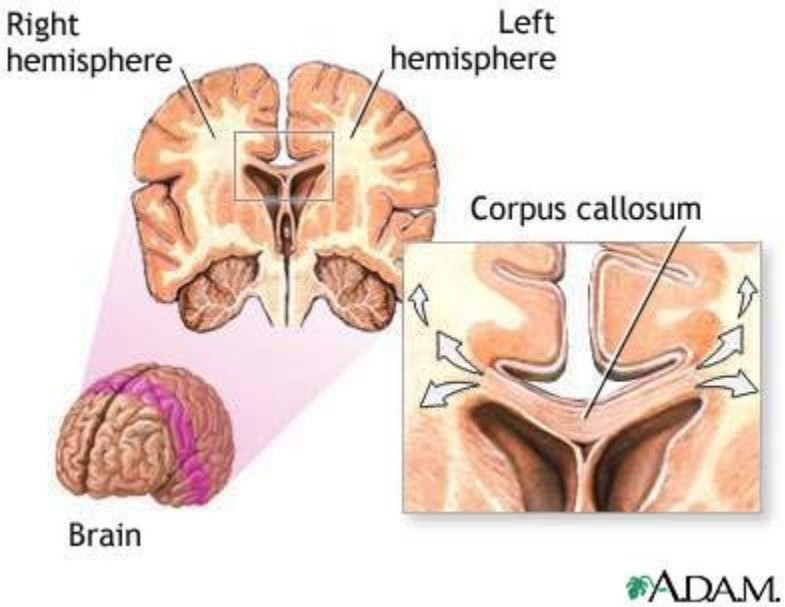
Major Landmarks of the Brain

- Two **<u>hemispheres</u>** are separated by the <u>longitudinal fissure</u>



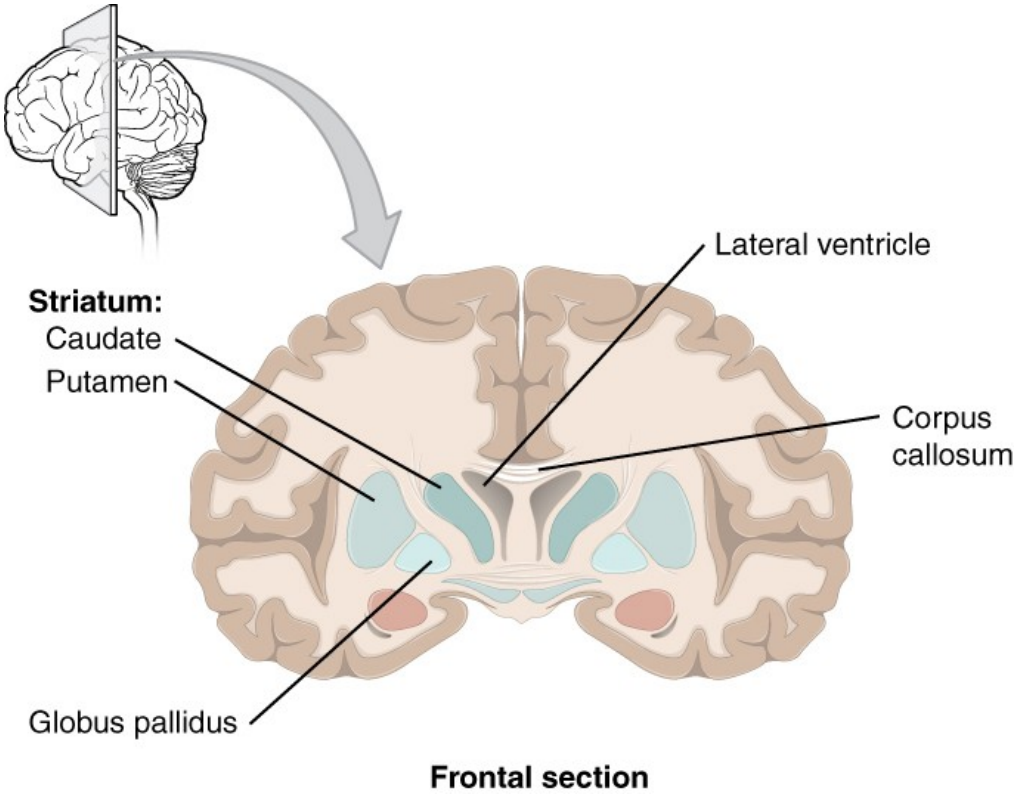
- Two hemispheres are connected by the **<u>corpus callosum</u>**, which allows communication between them



The Cells of the Brain are Organized into Different Structures

- **<u>Cortex</u>**—layers of nerve cells on the outer part of the brain

- **<u>Nuclei</u>**—clusters of nerve cells within the brain

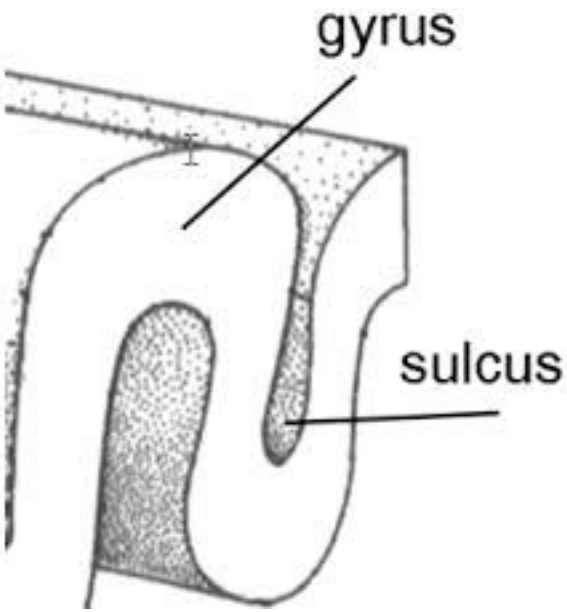


Folds of the Cortex

- **<u>Gyri</u>** (singular gyrus)– bulges or ridges on brain's surface

- **<u>Sulci</u>** (singular sulcus)— grooves or indentations into the brain's surface

Why is the surface of the brain so folded? What does this do?

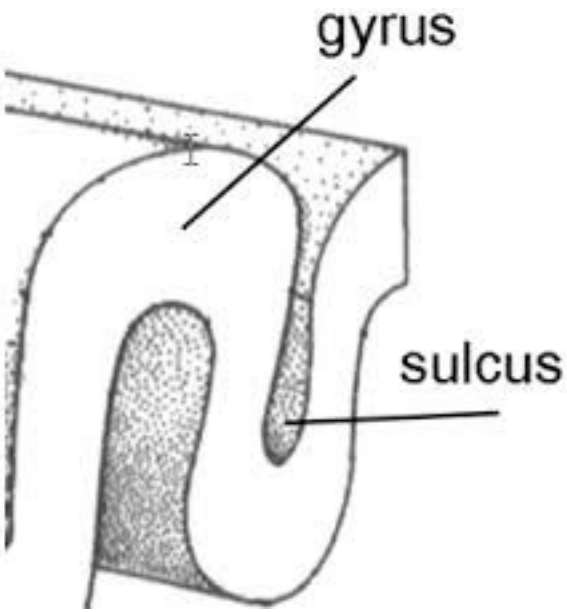


Folds of the Cortex

- **<u>Gyri</u>** (singular gyrus)– bulges or ridges on brain's surface

- **<u>Sulci</u>** (singular sulcus)— grooves or indentations into the brain's surface

Folding of the brain's surface greatly increases its surface area

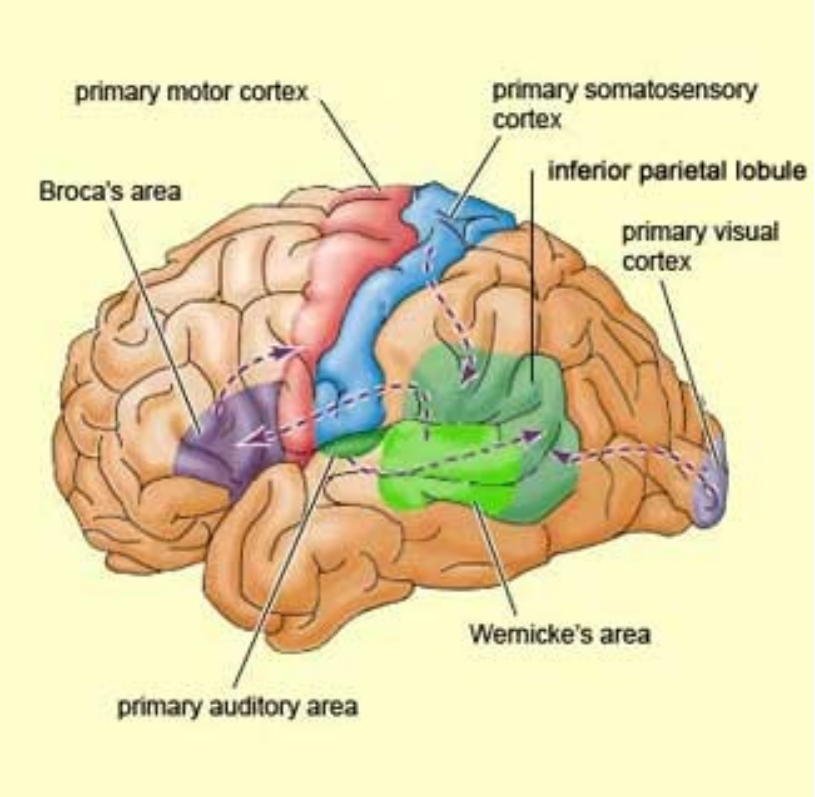


The Hemispheres are Both Symmetrical and Asymmetrical

- **Symmetrical**—structures found in both hemispheres of the brain—<u>sensory areas, motor areas</u>

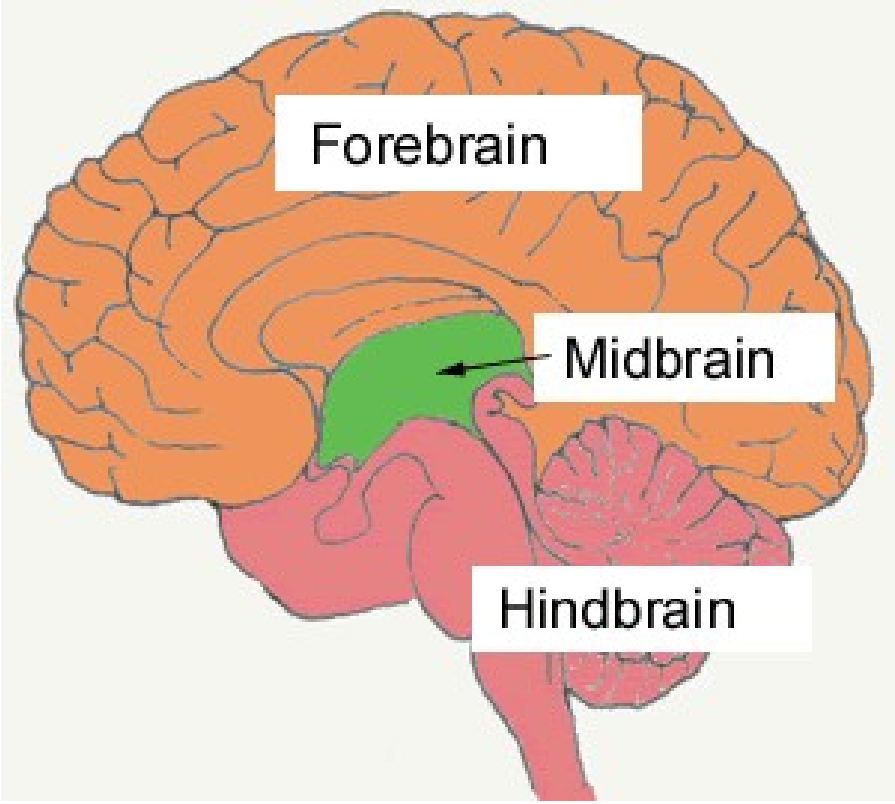
- **Asymmetrical—**structures found in one hemisphere only--<u>language</u> is located on the left and <u>spatial navigation</u> areas are located on the right in most people

- **Contralateral**—sensory information is sent to sensory areas and motor commands are sent from the motor areas on the opposite side of the body



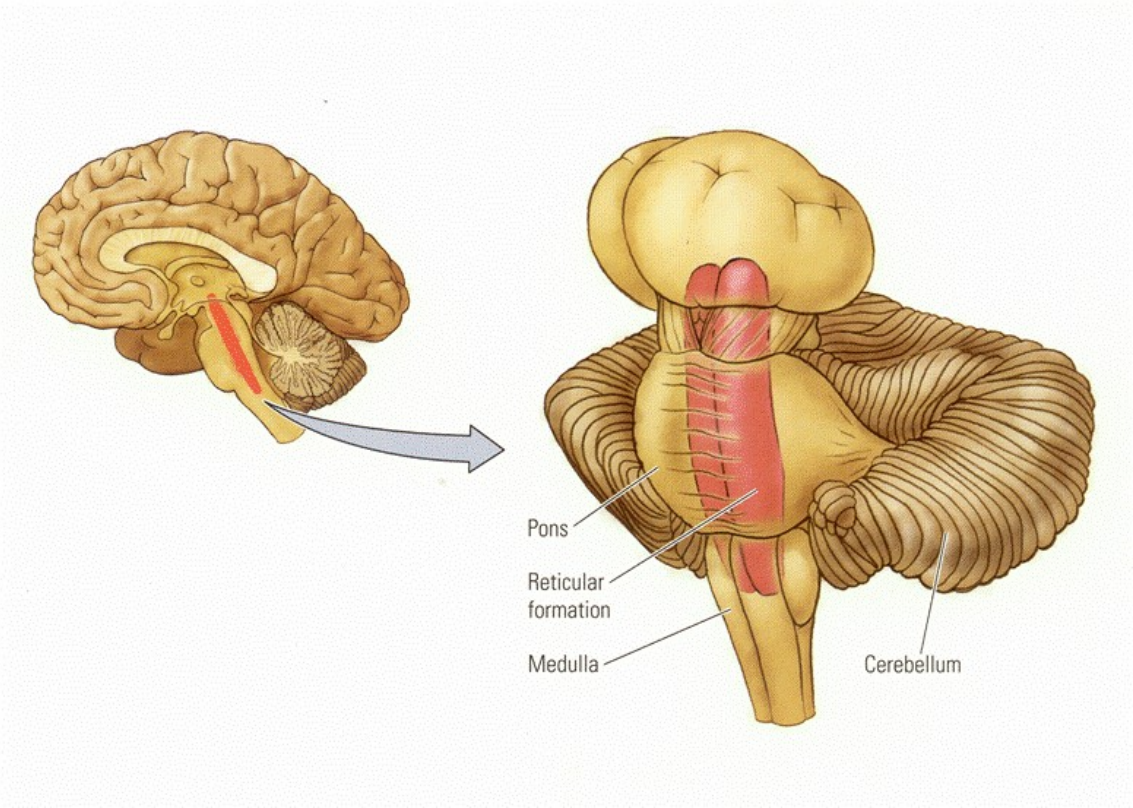
3 Main Subdivisions of the Brain

- Hindbrain
- Midbrain
- Forebrain



Functions of the Hindbrain

- Regulation of heart rate, breathing, blood pressure, swallowing
- Control of sleep and arousal
- Regulation of posture, movement and balance



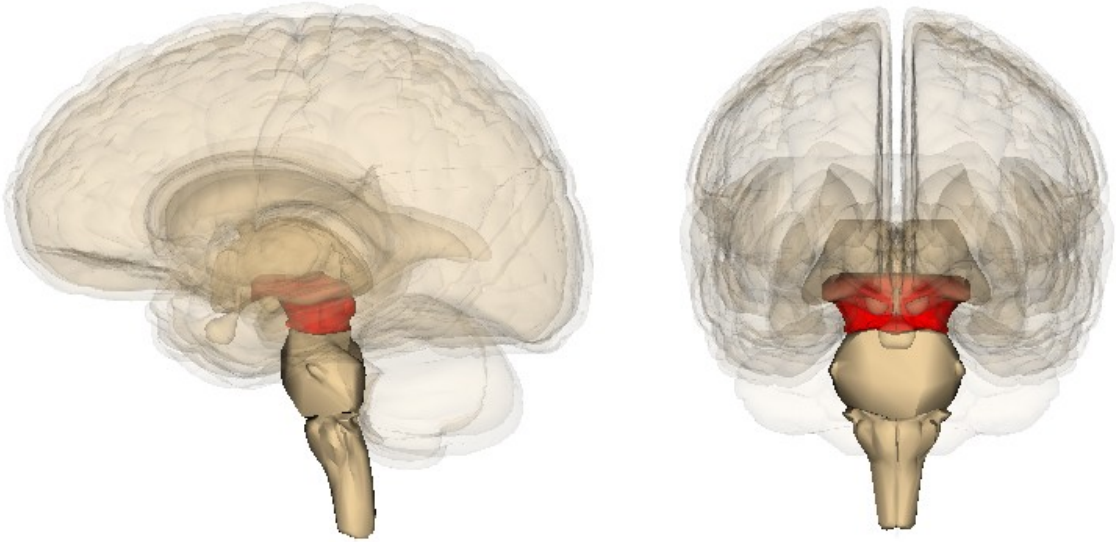
Key Point:

The hindbrain regulates functions that are basic and critical to life

Main Functions of the Midbrain

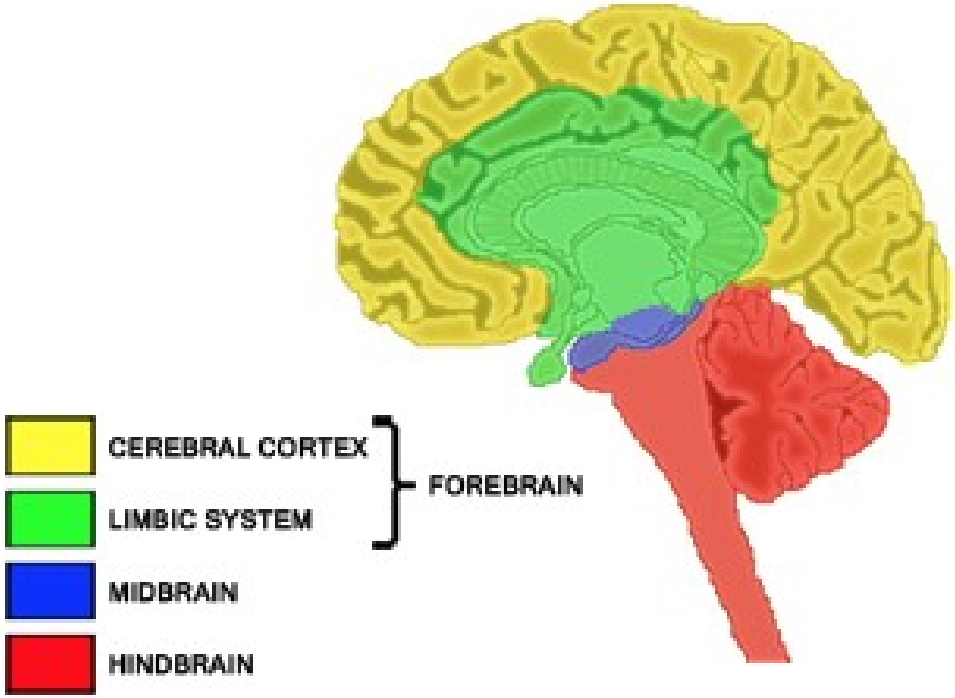
- Modulates movement, reward, and arousal

- Auditory and visual reflexes



Main Functions of the Forebrain

- Controls body temperature, hunger, thirst, and sleep
- Relaying and processing of sensory information
- Initiation and modulation of movement behaviors
- Generation of emotion
- Memory storage
- Cognition—thought, impulse control, planning

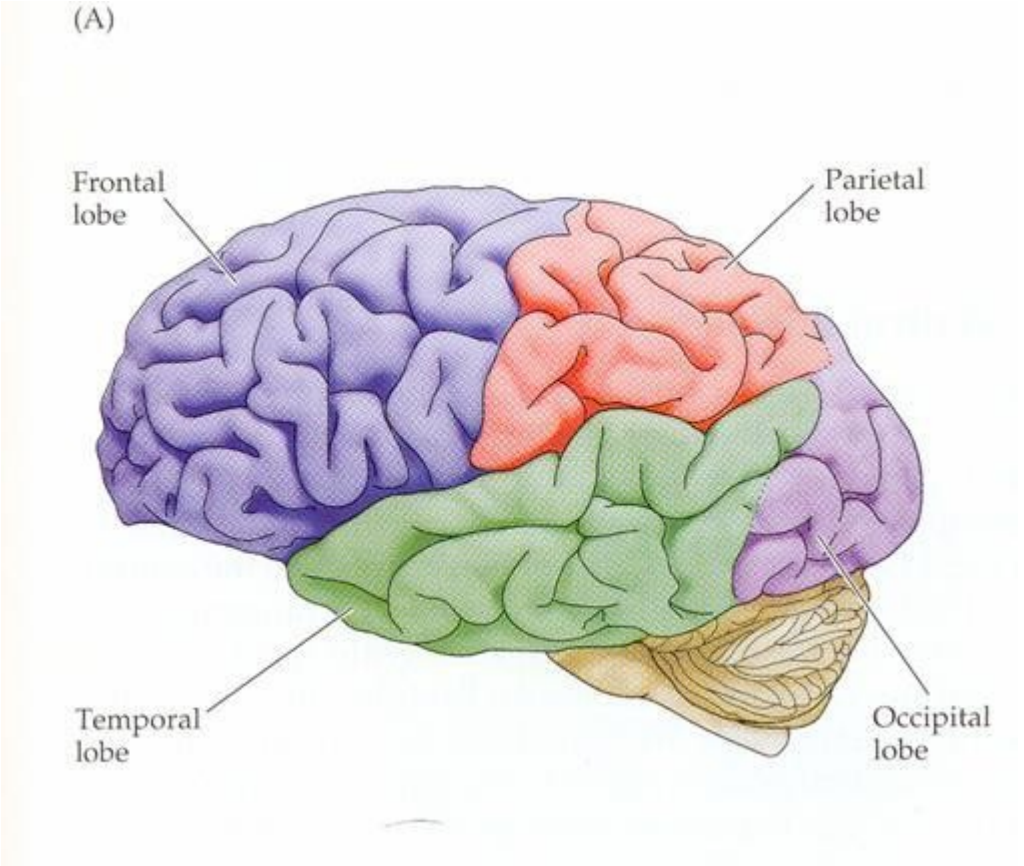


Key Point:

Many of the functions of the forebrain are what make us uniquely human—conscious thought, memory, emotion, etc.

Lobes of the Cerebral Cortex

- **Occipital**—back of the head
- **Temporal**—lower sides of the head
- **Parietal**—upper sides of head
- **Frontal**—front of head

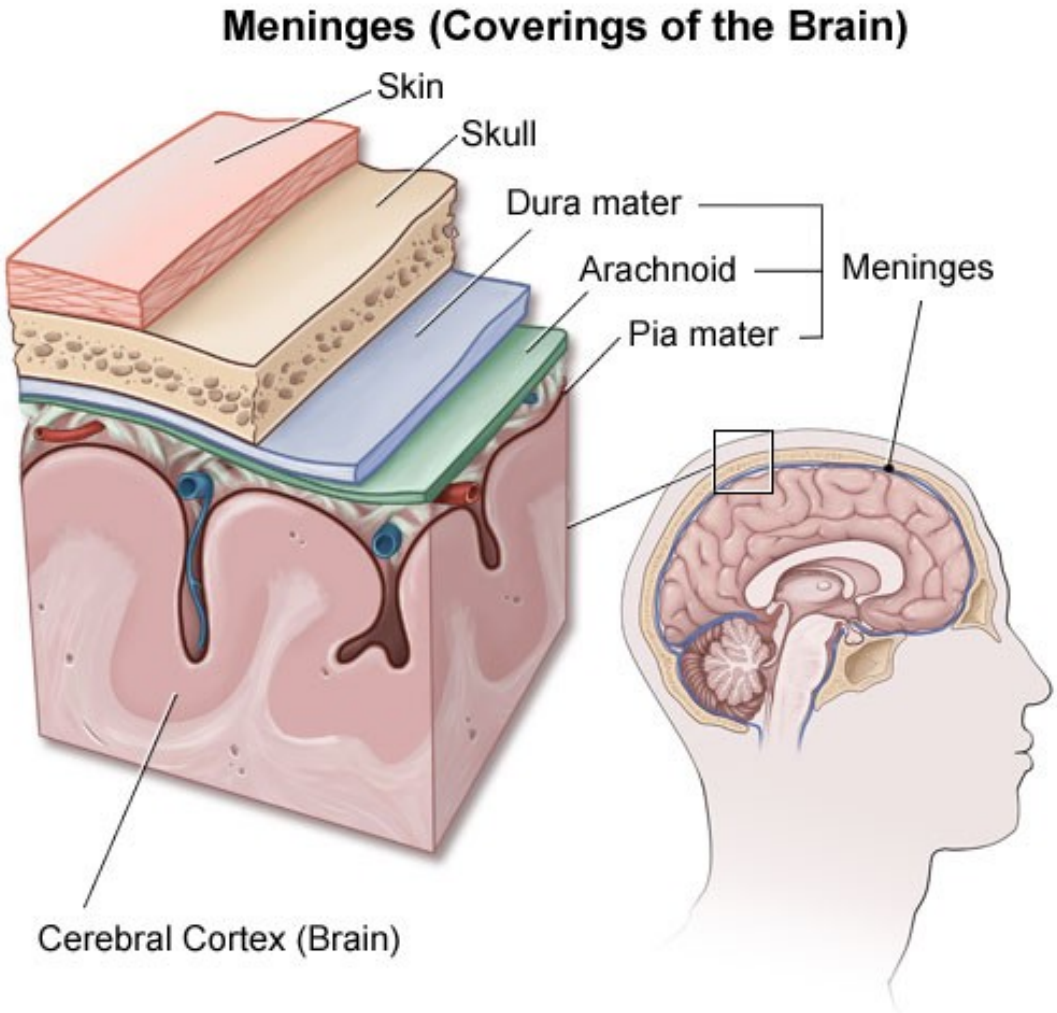


The cerebral cortex is the largest part of the brain in humans and is much larger than in other animals

Other parts of the brain

Meninges—Membranes Surrounding & Protecting the Brain

- **Dura mater**—outer-most later, very tough
- **Arachnoid membrane**—spiderweb-like middle layer
- **Pia mater**—adheres to brain and spinal cord



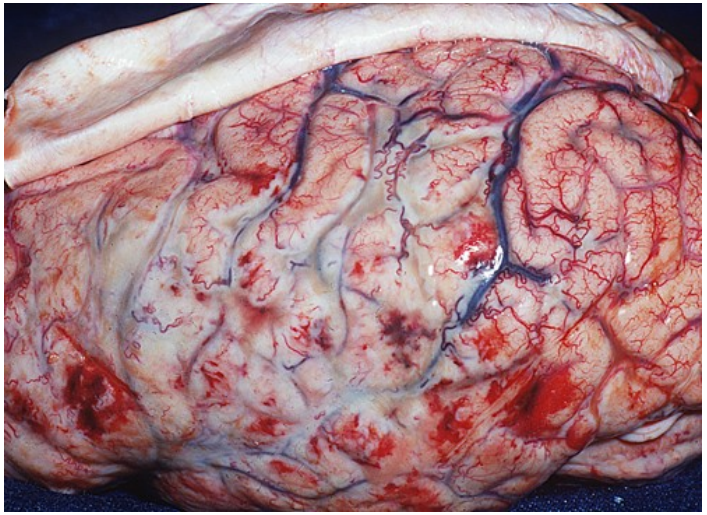
Diseases of the Meninges

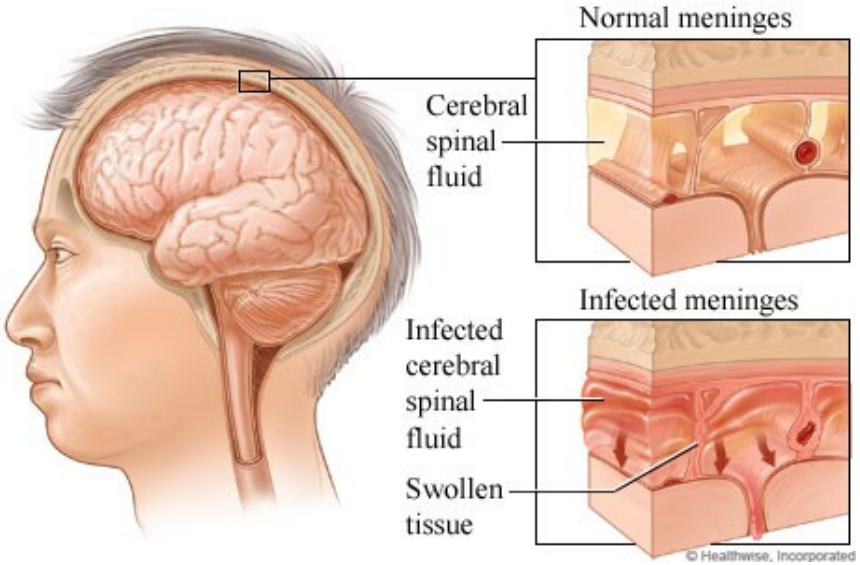
- **Meningitis**—infection or inflammation of the protective membranes surrounding the brain

- Can be viral or bacterial; viral has no cure

- The most common symptoms of meningitis are headache and neck stiffness associated with fever, confusion or altered consciousness, vomiting, and an inability to tolerate light (**photophobia**) or loud noises (**phonophobia**)

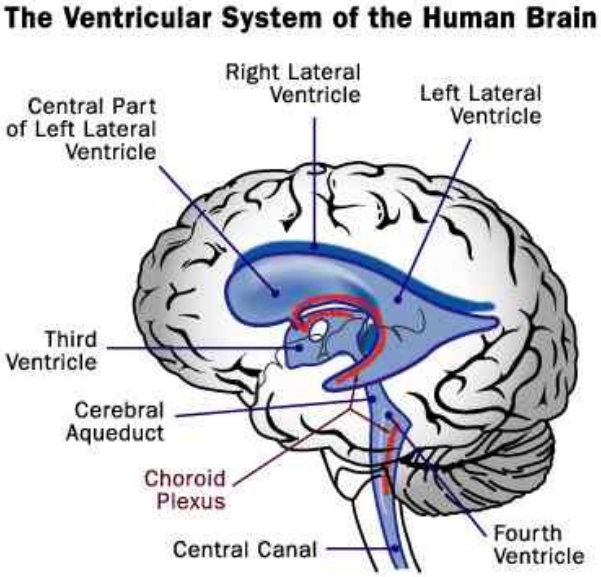
- Approximately 40,000 people get meningitis each year in the U.S.

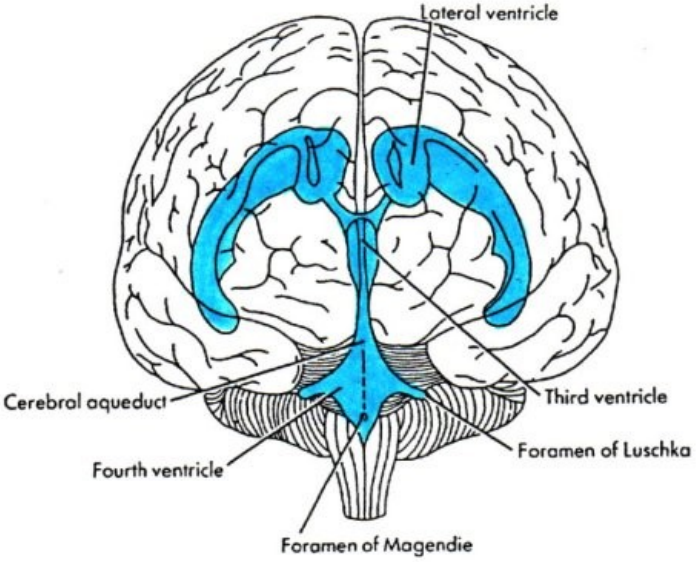




Ventricles—Large Fluid-Filled Chambers in the Brain

- 4 ventricles—two lateral, third, and fourth
- Contain cerebrospinal fluid (CSF)
- CSF is produced by **<u>choroid plexuses</u>** and trickles out of ventricles and washes through the brain into the circulation

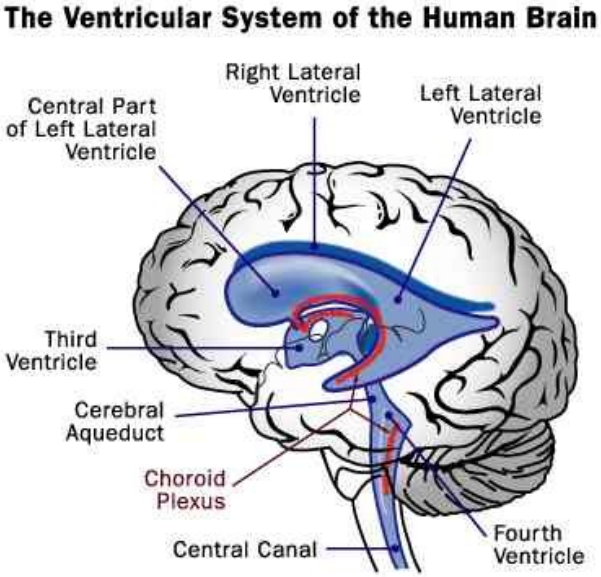


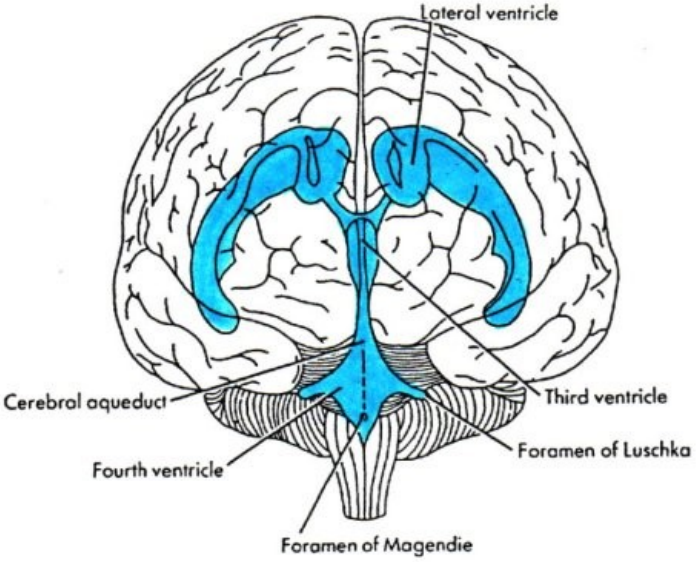


What are the functions of the ventricles and cerebrospinal fluid?

Ventricles—Large Fluid-Filled Chambers in the Brain

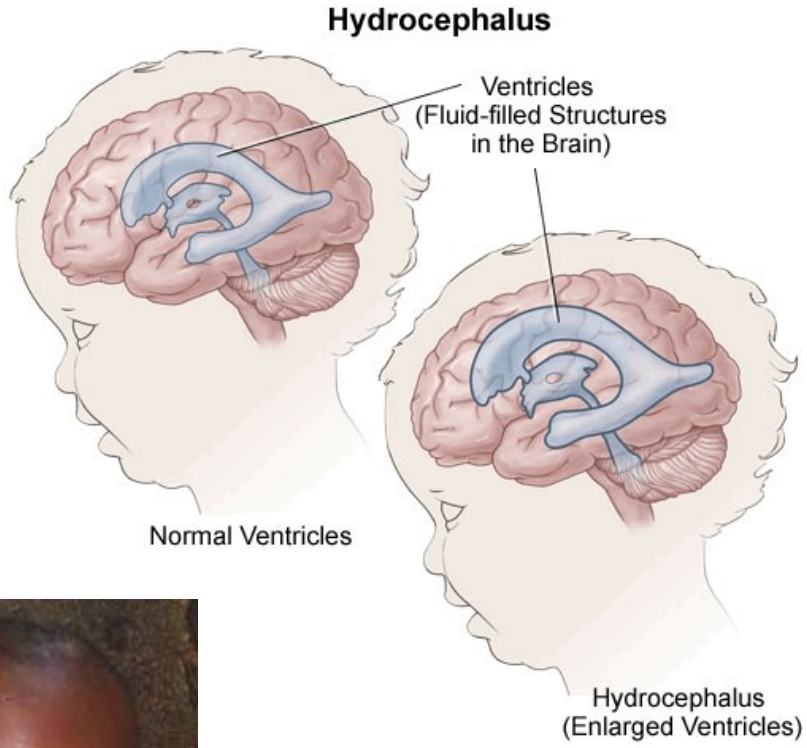
- 4 ventricles—two lateral, third, and fourth
- Contain cerebrospinal fluid (CSF)
- CSF is produced by **choroid plexuses** and trickles out of ventricles and washes through the brain into the circulation
- CSF-filled ventricles cushion the brain from shocks
- CSF also flows through **glymphatic system** to flush toxins out of brain



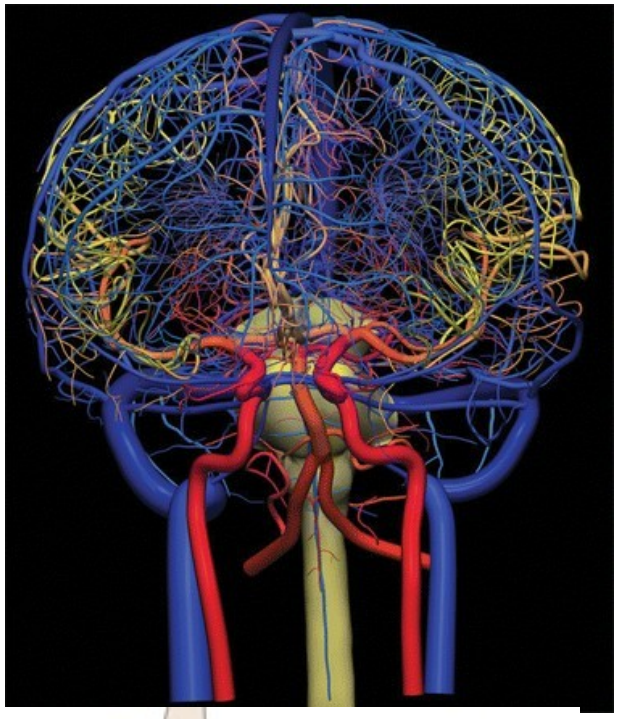


Hydrocephalus

- **<u>Hydrocephalus</u>**— blockage of the drainage of CSF through the brain, causing the ventricles to swell
- Can cause brain damage if not reversed
- Symptoms are obvious and excessive head size and "**<u>sundowning</u>**", deviation of the eyes downward due to high intracranial pressure
- Medical shunts can help

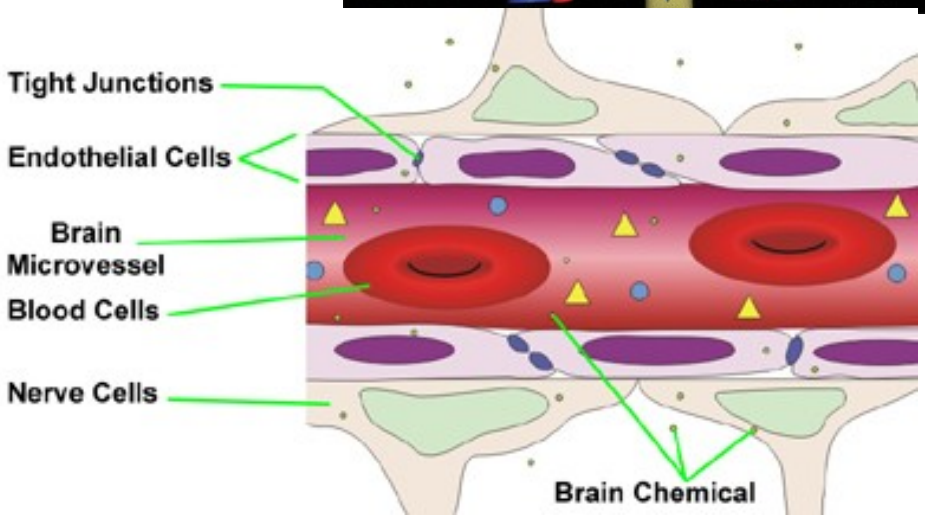


The Cerebral Vasculature and the Blood-Brain Barrier



- **<u>Cerebral vasculature</u>**—blood vessels supplying the brain with oxygen and nutrients

- **<u>Blood-brain barrier</u>**—tight seals and other barriers around blood vessels that restrict movement from the blood into the brain and vice versa

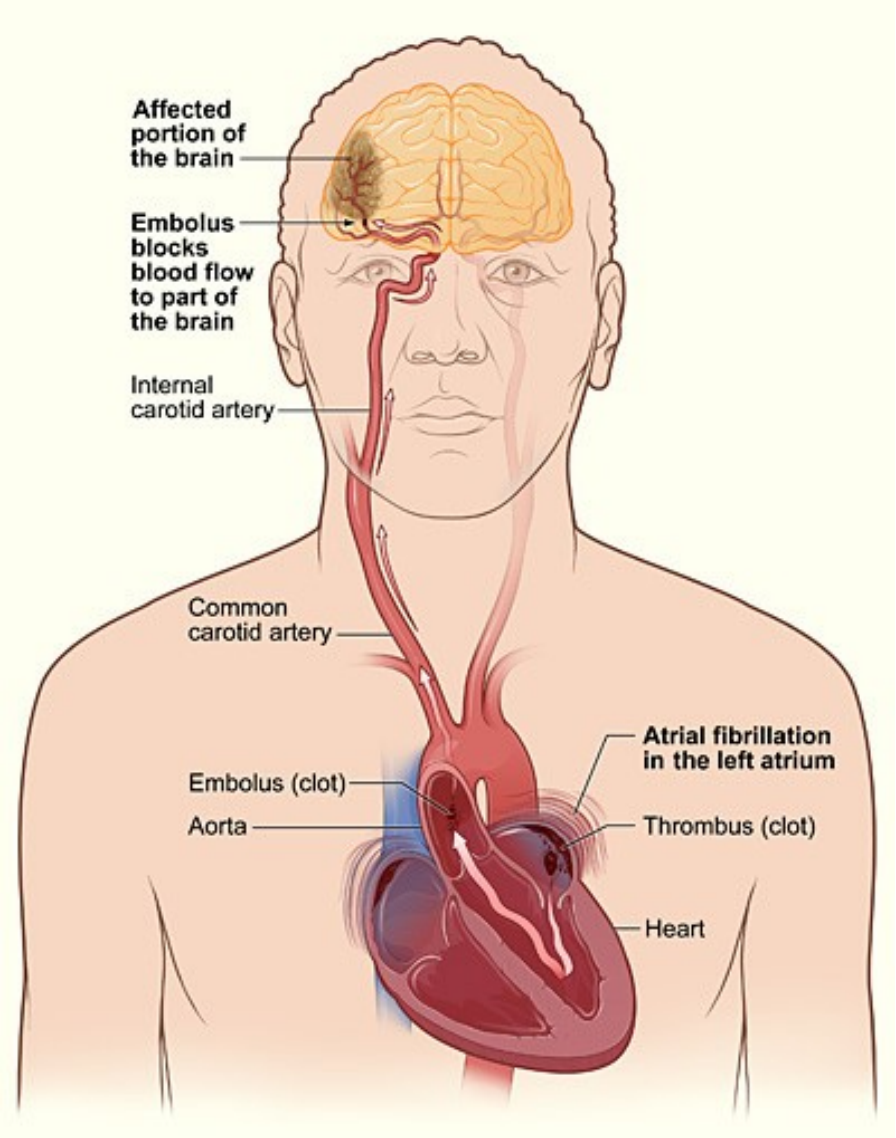


Stroke

- **Stroke**—Decrease in blood supply providing oxygen to a part of the brain that results of death of neurons downstream of the blockage

- Stokes often happen <u>unilaterally</u>—to one side of the brain only—and often have affects on sensory (numbness) and motor (paralysis) function on one side of the body

- Depending on where it happens it can adversely affect sensorimotor information, language, spatial navigation, etc.

- **<u>Aphasia</u>**—loss of ability to understand or make language, often occurs with stroke

- More than 795,000 people have a stroke each year in America